BA ZC581 :
Applied Data Science
Module: 11

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Disclaimer

- The slides presented here are obtained / extracted from the authors of the prescribed books and from various other subject contributors.
- I hereby thank and acknowledge all the contributors for their material and inputs.
- I have added and modified a few slides to suit the requirements of the course.

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Module 11:
EDA: Statistical Summaries

Module 10 Topics

Topic 1: Descriptive Statistics

Topic 2: Correlations

Topic 3: Skewness & Kurtosis

Topic 4: Business Insight Extraction

Topic 5: Demos

Ref: T1 , Ch 4

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Topic 1: Descriptive Statistics

Descriptive Statistics

- Summarizes and describes the main features / patterns of a dataset in a simple meaningful way.
- The first stage in Data Analysis
- Does not predict, but helps one see how the data looks like
- Essentially
  - Organizes data
  - Highlights common patterns
  - Provides a snapshot of core properties of data

Why it matters?

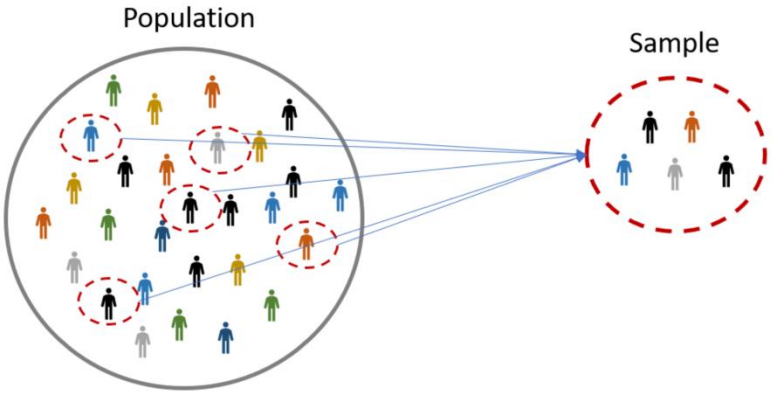
- Allows teams to quickly answer questions like
  - What is our average sales per day?
  - How much do customer's ratings vary
  - What is the typical delivery time?
  - What is the percentage of items returned (quantity and value)?
  - Are there any unusual spikes or dips

Statistical Analysis

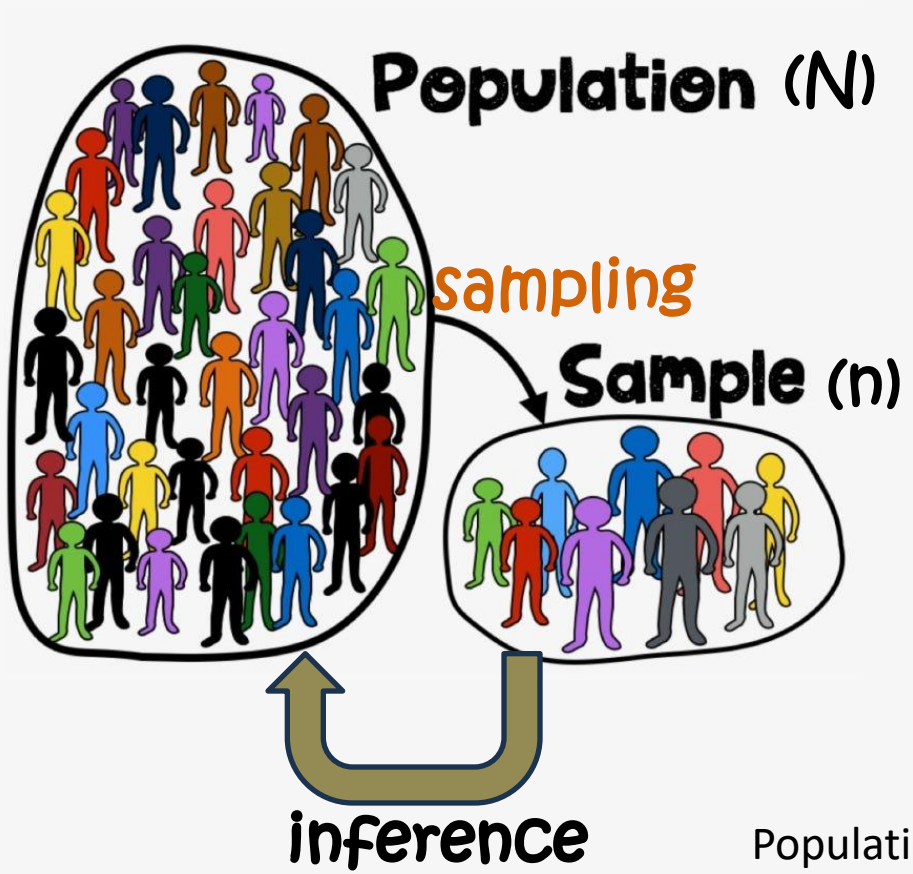
- Used to gain an understanding of a larger set of data, called then population!
- Inferential statistics is often defined as the science of drawing conclusions about a population from observations made on a representative sample of that population.
- It allows inferences to be drawn about target markets and the general population by expanding findings to predict the behaviour of the entire population based on the few items sampled.

Population & Sample

- **Population** is the entire group that we wish analyse and draw conclusions about.
- **Sample** is the specific group of individual elements taken from the population that represent the population in the context of the area of interest of analysis.



Population & Sample



Population subset is the sample

Samples

- The population can be defined in terms of geographical location, age, income, etc.
- If population is large and mixed, the representative sample may be difficult to obtain
- Samples should be a true representation in the context of the analysis.
  - If the analysis is on a location, sample should contain representation from all locations
- **Inferences on the samples are extended on to the population**

Why Sampling?

- Populations are too large to study fully
- Saves time and money
- Allows faster decision making
- Reduces effort without reducing accuracy
- Population could be increasing very rapidly
- Enables quality control
- It is used when it is impossible to study the entire population, or to

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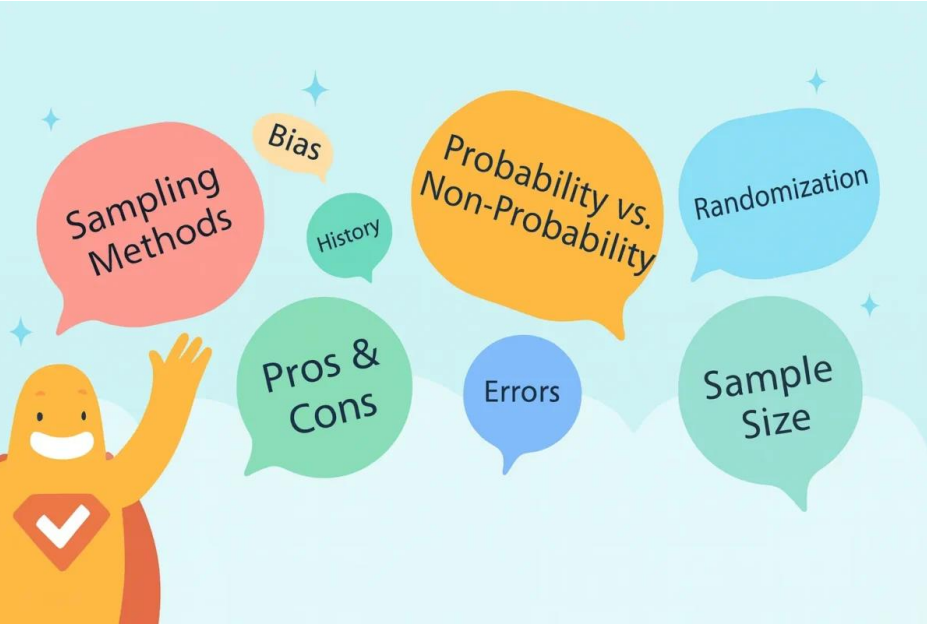
Sampling Techniques

Sampling Types

- Probabilistic Sampling:
  - Every member of the population has a chance of being selected.
  - Used in quantitative research.
  - Essentially representative of the whole population.
- Non-probabilistic Sampling
  - Samples are selected based on non-random criteria, and not every individual has a chance of being included in the sample set.
  - Higher risk of sampling bias
  - Used in qualitative / exploratory research

Probabilistic Sampling



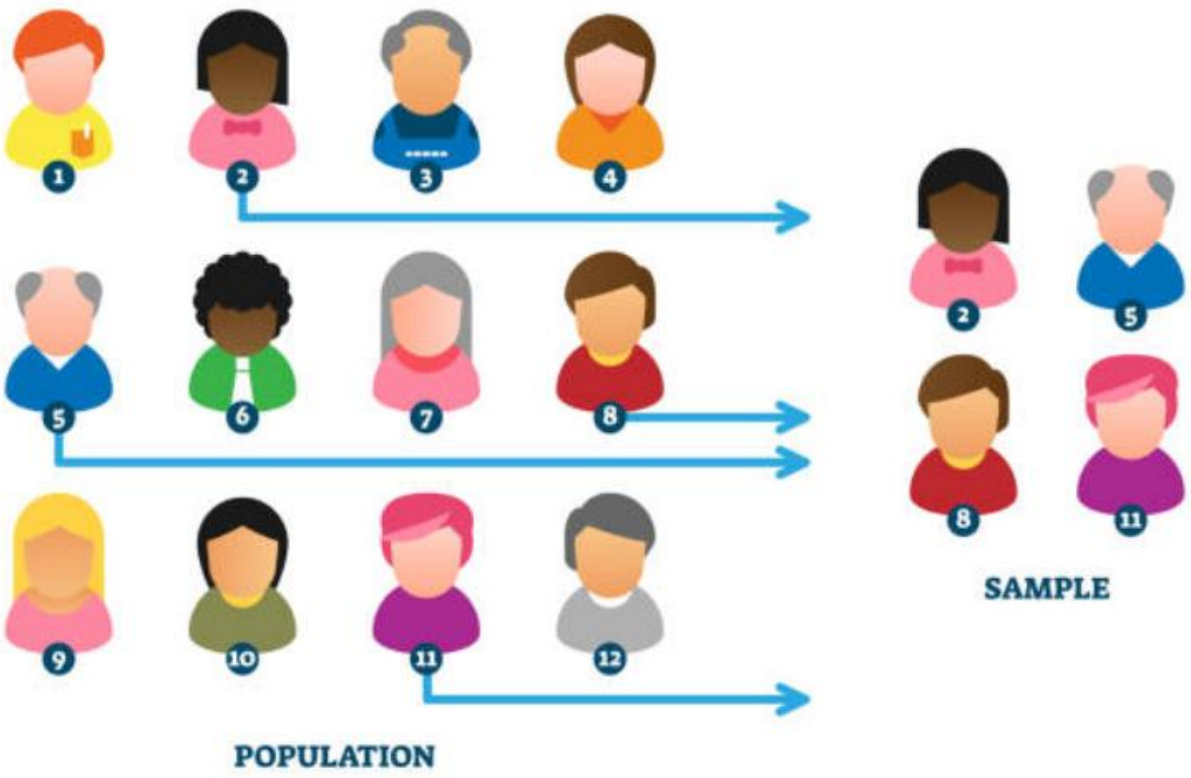
- Random Sampling
- Systematic Sampling
- Stratified Sampling
- Clustered Sampling

Random Sampling

- Steps:
  - Prepare a list of all population members
  - Assign a unique Id to each member element (if the population has N elements, assign an Id from 1 to N)
  - Use a random number generator to generate S (sample size) unique numbers between 1..N
  - Select the samples for the random Ids generated

Random Sampling

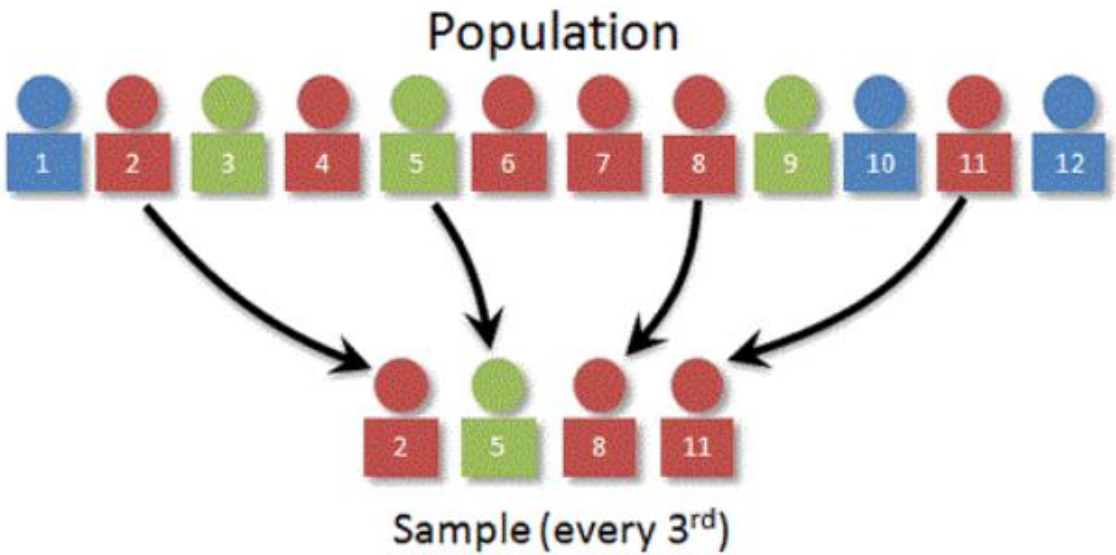
Every element has an equal probability of being selected

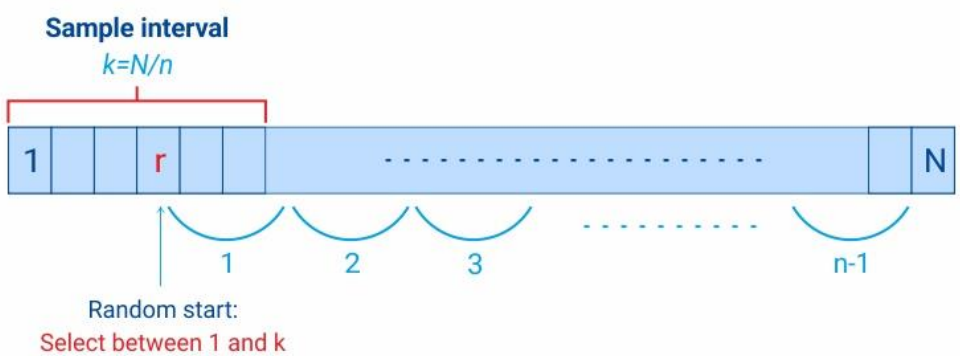


Systematic Sampling

- Also called Linear Sampling
- Steps
  - Prepare a list of population members
  - Assign a unique Id for every member (Id will be 1 to N for N elements in the population)
  - Identify the sample size S
  - Calculate the interval (standard distance between intervals)
    - Interval i = N / S
  - Select the 1st sample at position r (it could be selected randomly, or the first one in the list)
  - Select samples at regular intervals (i) from this 1st sample (r, r+i, r+2*i, r+3*i …)

Systematic Sampling

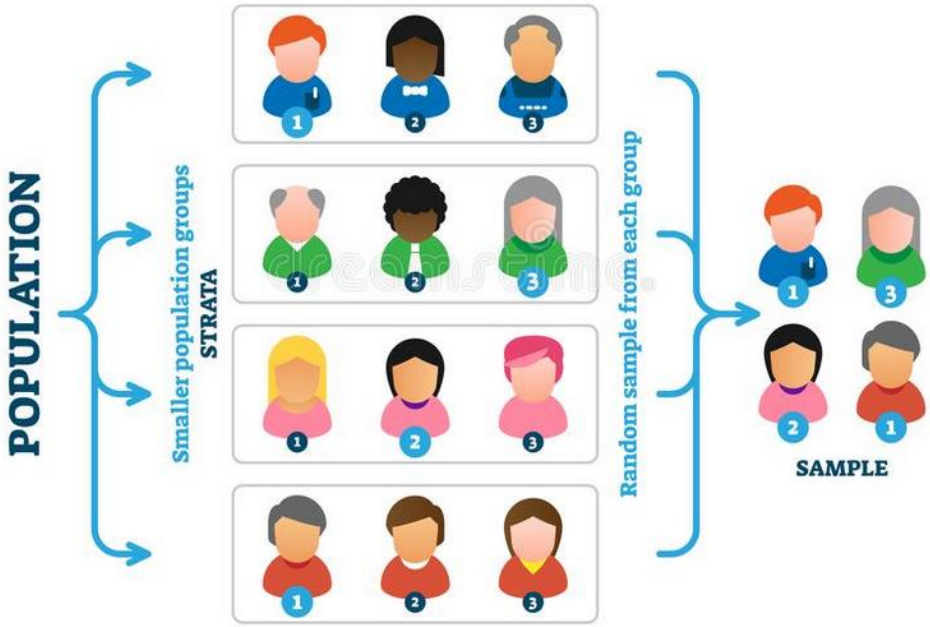




Stratified Sampling

- Every strata has a representation
- Steps:
  - Identify the stratification variable and the number of strata to be used. (classes in data)
    - Considering the entire population, strata should be covering every element
    - Differences between elements in the strata should be minimum
    - Each strata should be extremely different from the other
  - Assign Ids to each member in each stratum
  - Select random elements from each stratum to form the sample
    - Proportional or disproportional to the size of stratum

Stratified Sampling



Proportionate Stratified Random Sampling Formula: $s_h = (S_h / N) * S$

s_h = sample size for h^{th} stratum

S_h = population size of hth stratum

N = population size

S= sample size

Stratified Sampling

Proportionate Stratified Sampling

Stratum	A	B	C	D
Population Size	500	1000	2000	1500
Sampling Fraction	0.1	0.1	0.1	0.1
Samples	50	100	200	150

Disproportionate Stratified Sampling

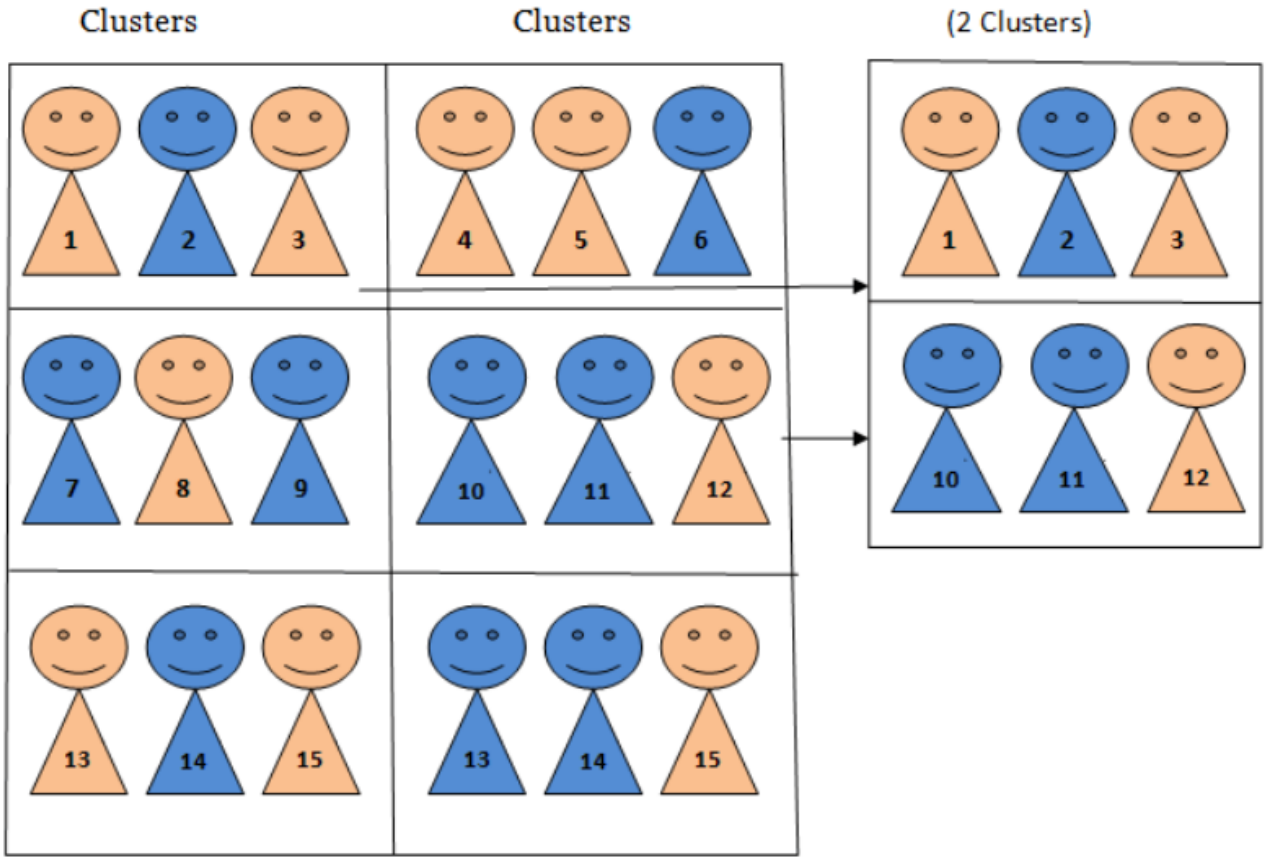
Stratum	A	B	C	D
Population Size	500	1000	2000	1500
Sampling Fraction	0.25	0.15	0.1	0.05
Samples	125	150	200	75

Clustered Random Sampling

- Steps
  - Decide the target audience and sample size
  - Determine the groups (may be same average number in each group) having similar properties
    - Groups are clusters, every member is similar to the others in the group, and very different from members in other groups
  - Select complete clusters randomly
  - Create sub types to allow 2 level and multi level clusters
  - If clusters are large, sample individuals from each cluster using other sampling techniques. This leads to a multi stage clustered sampling

Clustered Random Sampling

Clustered Random Sampling



Sample Size

- It is the number of subjects included in the sample set selected from the population.
- Considered an adequate number that represents the real population for the specific analysis
  - How certain we want to be that the sample reflects the population (90%, 95%, 99%)
  - Confidence level determines error margin in Z score to be used

$$\text{Sample Size} = \frac{(\text{Z-score})^2 \times \text{StdDev} \times (1-\text{StdDev})}{(\text{confidence interval})^2}$$

Sample Size Determination

Confidence level	z-score
80%	1.28
85%	1.44
90%	1.65
95%	1.96
99%	2.58

Say you choose to work with a 95% confidence level, a standard deviation of 0.5, and a confidence interval (margin of error) of ± 5% (= 0.05), you just need to substitute the values in the formula:

((1.96)^2 x 0.5(1-0.5)) / (.05)^2

(3.8416 x .25) / .0025

.9604 / .0025

384.16

The sample size should be 385.

Sample Size Determination

- Apply Finite population correction if N is known.

$$n = \frac{n_0}{1 + \frac{n_0 - 1}{N}}$$

- Here n0 is the sample size computed earlier and N is the population size

$$n = \frac{384.16}{1 + \frac{384.16 - 1}{10000}}$$

$$n \approx 370$$

- Hence, n = 370

Sample Size

- Thumb Rule:
- **A good maximum sample size is usually around 10% of the population, as long as this does not exceed 1000**.
  - For example, in a population of 5000, 10% would be 500.
  - In a population of 200,000, 10% would be 20,000. This exceeds 1000, so in this case the maximum would be 1000.

Statistical Analysis

- Common Descriptive statistics
  - Mean
  - Median
  - Variance
  - Std Deviation
  - Range
  - Percentile
- Based on relationships
  - Covariance
  - Coorelation
  - etc

Measure of Central Tendencies

Measures of central tendency describe where most of the values in a data set lie. Three main measures of central tendency are often reported:

Mode: the most popular response or a value that occurs most often in the data set.

Median: the value in the exact middle of the data set when ordered from low to high.

Mean: the sum of all values divided by the number of values. (average)

Other measures of variability

Measures of variability indicate how spread out the values in a data set are. Four main measures of variability are often reported:

Range: the highest value minus the lowest value of the data set.

Interquartile range (IQR): the range of the middle half of the data set.

Standard deviation: the average distance between each value in your data set and the mean.

Variance: the square of the standard deviation.

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Topic 2: Correlation

Relationship between Data

- Establish means of evaluating the relationship (dependency) of one variable over the other
- Measures of relationship
  - **Co variance**
  - **Correlation**
  - **Regression (we will study later)**
- We use **Covariance** to measure how much two variables change with each other.
- **Correlation** reveals the *relation* between the variables. We use correlation to determine how strongly two variables are linked to each other.
  - Ex: if the share price of Reliance increases, will the share price of Infosys increase, fall or not impacted ?

Covariance

- **Covariance** is a measure of how much two random variables vary together.
- When one variable changes, does the other tend to change in the same direction, the opposite direction, or not at all?
- It is similar to variance,
  - **variance** tells you how a ***single*** variable varies,
  - **co variance** tells you how **two** variables vary together.

Covariance

for population

$$\mathrm{Cov}_{\mathrm{population}}(X,Y) = \frac{1}{N}\sum(x_i - \mu_X)(y_i - \mu_Y)$$

for sample

$$\mathrm{Cov}_{\mathrm{sample}}(X,Y) = \frac{1}{N-1}\sum(x_i - \bar{X})(y_i - \bar{Y})$$

Bessel correction

- Interpreting covariance is hard to gain any meaning from; since the values are not scale dependent (ex: pounds and inches)
- It is not a normalised measure and does not have any upper bound.

Interpret Covariance

1. Sign identifies the direction of the relationship
   1. Positive: move together in same direction
   2. Negative: move in the opposite direction
   3. Near zero: variables do not have any relationship
2. Magnitude is not interpretable. It depends on scale of the data and the data spread.
3. Use correlation when we need the strength of the relationship

Example

x: 2.1, 2.5, 3.6, 4.0 (mean μ = 3.1)

y: 8, 10, 12, 14 (mean v = 11)

$$Cov(X,Y) = \Sigma E((X-\mu)(Y-v)) / (n-1)$$

$$= ((2.1-3.1)(8-11)+(2.5-3.1)(10-11)+(3.6-3.1)(12-11)+(4.0-3.1)(14-11)) /(4-1)$$

$$= ((-1)(-3) + (-0.6)(-1)+(.5)(1)+(0.9)(3)) / 3$$

$$= (3 + 0.6 + .5 + 2.7) / 3$$

$$= 6.8/3$$

$$= 2.267$$

Correlation

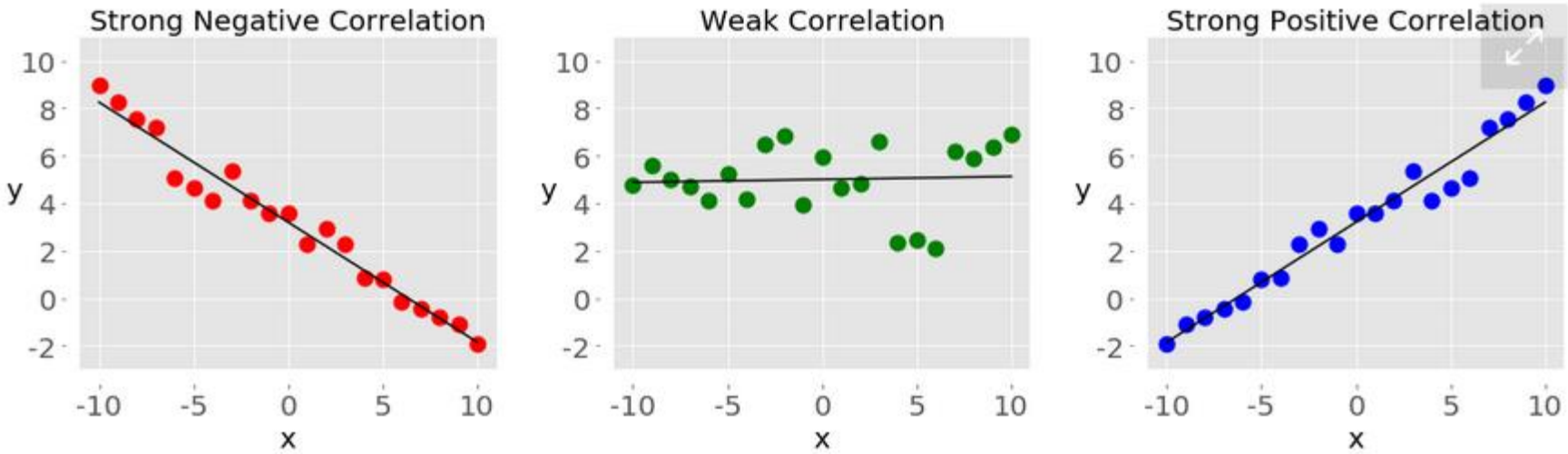
- Correlation is **standardized covariation**.
- It takes covariance and divides it by the product of the standard deviations:

$$\text{Correlation} = \frac{\text{Cov}(X, Y)}{\sigma_X \cdot \sigma_Y}$$

- It is unit-free
- It always lies between −1 and +1
- So correlation tells you:
  - Direction (same as covariance)
  - Strength (because it's standardized)

Correlation

- Identifies a relationship between two attributes



- Usually defined as a value between -1 to +1 (value 0 indicates No correlation)

Correlation Coefficient

- The correlation coefficient is a value that indicates the strength of the relationship between variables. The coefficient can take any values from -1 to 1. The interpretations of the values are:
  - **-1:** Perfect negative correlation. The variables tend to move in opposite directions (i.e., when one variable increases, the other variable decreases).
  - **0:** No correlation. The variables do not have a relationship with each other.
  - **+1:** Perfect positive correlation. The variables tend to move in the same direction (i.e., when one variable increases, the other variable also increases).

Correlation and causation

- If two variables are correlated, it does not imply that one variable causes the changes in another variable.
- Correlation only assesses relationships between variables, and there may be different factors that lead to the relationships.
- Causation may be a reason for the correlation, but it is not the only possible explanation.

Direct Method

$$r_{xy} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$$

r_{xy} – the correlation coefficient of the linear relationship between the variables x and y

x_i – the values of the x-variable in a sample

$\bar{x}$ – the mean of the values of the x-variable

y_i – the values of the y-variable in a sample

$\bar{y}$ – the mean of the values of the y-variable

Pearson's correlation

- Pearson's correlation coefficient is the ratio of the covariance of the two variables to the product of their standard deviations.
- Value in the range -1 to + 1

$$r_{xy} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

where:

- n is sample size
- x_i, y_i are the individual sample points indexed with *i*
- $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$ (the sample mean); and analogously for $\bar{y}$

Example:

Assume we have 2 variables :

X : hours studied; Y: exam rank (1= best)

Student	X (Hours)	Y (Rank)
1	2	9
2	3	7
3	4	6
4	5	5
5	6	4
6	7	3
7	8	2
8	9	1
9	3	8
10	4	7

Example:

Step 1: Compute means

$$\bar{X} = \frac{2+3+4+5+6+7+8+9+3+4}{10} = 5.1$$

$$\bar{Y} = \frac{9+7+6+5+4+3+2+1+8+7}{10} = 5.2$$

Step 2: Compute deviations, products and squares

X	Y	X−X̄	Y−Ȳ	(X−X̄)(Y−Ȳ)	(X−X̄)²	(Y−Ȳ)²
2	9	-3.1	3.8	-11.78	9.61	14.44
3	7	-2.1	1.8	-3.78	4.41	3.24
4	6	-1.1	0.8	-0.88	1.21	0.64
5	5	-0.1	-0.2	0.02	0.01	0.04
6	4	0.9	-1.2	-1.08	0.81	1.44
7	3	1.9	-2.2	-4.18	3.61	4.84
8	2	2.9	-3.2	-9.28	8.41	10.24
9	1	3.9	-4.2	-16.38	15.21	17.64
3	8	-2.1	2.8	-5.88	4.41	7.84
4	7	-1.1	1.8	-1.98	1.21	3.24

Sums:

$$\sum (X-\bar{X})(Y-\bar{Y}) = -55.2$$

$$\sum (X-\bar{X})^2 = 48.9$$

$$\sum (Y-\bar{Y})^2 = 63.6$$

Example:

Step 3: Apply Pearson's formulae

$$r = \frac{\sum(X-\bar{X})(Y-\bar{Y})}{\sqrt{\sum(X-\bar{X})^2 \cdot \sum(Y-\bar{Y})^2}}$$

$$r = \frac{-55.2}{\sqrt{48.9 \cdot 63.6}}$$

$$r = \frac{-55.2}{\sqrt{3107.04}}$$

$$r = \frac{-55.2}{55.74}$$

$$r = -0.99$$

Step 4: Interpretation

- Extremely strong negative linear relationship
- As hours studied increase, exam rank improves (rank number decreases)
- The relationship is almost perfectly linear

Alternate raw form

Good for hand calculations:

$$r = \frac{n\sum xy - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2]\,[n\sum y^2 - (\sum y)^2]}}$$

Kendal's Correlation

- Is used to measure the ordinal association (rank order) between two measured quantities.
  - how consistently two variables move *up or down together* based on their ordering, not their numeric distance
- Denoted by Tau, value is in range -1 to +1
- Identifies the degree of concordance between two columns of ranked data of the samples

Example:

Assume we have 2 variables :

X : hours studied; Y: exam rank (1= best)

Student	X (Hours)	Y (Rank)
1	2	9
2	3	7
3	4	6
4	5	5
5	6	4
6	7	3
7	8	2
8	9	1
9	3	8
10	4	7

Example:

Step 1: Count concordant & Discordant pairs

A pair of points (x_i, y_i) and (x_j, y_j):

- **Concordant** if both X and Y move in the same direction (i.e., $x_i > x_j$ AND $y_i > y_j$, or both <)
- **Discordant** if X and Y move in opposite directions (i.e., $x_i > x_j$ AND $y_i < y_j$, or vice versa)

There are $\frac{10 \cdot 9}{2} = 45$ total pairs.

After comparing all 45 pairs:

- **Concordant pairs (C)** = 39
- **Discordant pairs (D)** = 6
- No ties in X or Y in this example

Example:

Step 2: Apply Kendal's Formulae

$$\tau = \frac{C - D}{\frac{n(n-1)}{2}}$$

$$\tau = \frac{39 - 6}{45}$$

$$\tau = \frac{33}{45} = 0.733$$

Step 3: Interpretation

- A Kendall's τ of **0.733** indicates:
  - A strong positive rank correlation
  - Students who study more hours tend to achieve better exam ranks
- The relationship is not perfect (because of the messy points), but still very strong

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Topic 3: Skewness & Kurtosis

Skewed Distribution

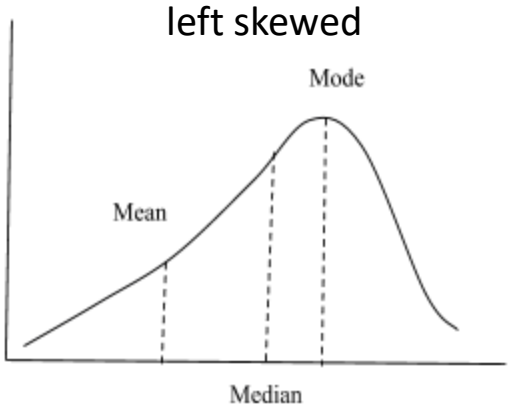
- A **skewed distribution** is typically asymmetric and has more values on one end than the other.
- It is a measure of lack of symmetry in the shape of the distribution

- Extreme outliers can also produce misleading statistics, so you may need a systematic approach to dealing with these values.

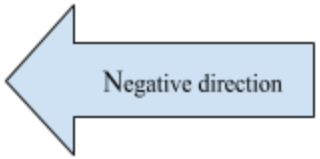
Positive & Negative Skew

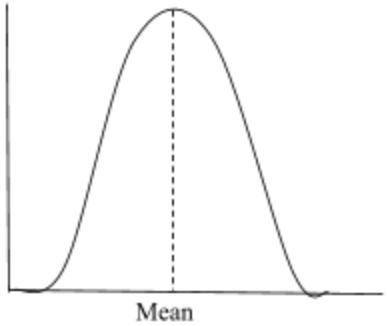
Q3-Q2 < Q2-Q1

left skewed



mean<median<mode

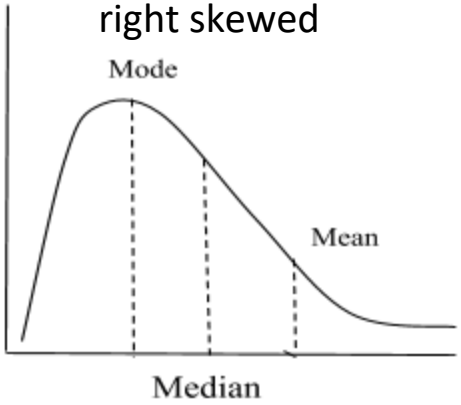




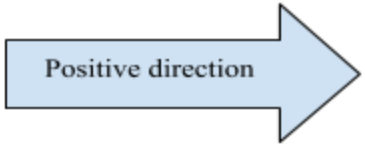
Symmetrical data
mean=median=mode

Q3-Q2 > Q2-Q1

right skewed



mode<median<mean



Good number of small values pull the mean down. Also we have many values more than median

Good number of high values pull the mean up. We have many values far less than median

Skewness Calculation

- Fisher-Pearson Coefficient of Skewness calculates the 3rd moment

$$\text{Skewness Formula} = \frac{\sum_i^N (X_i - \bar{X})^3}{(N-1) \times \sigma^3}$$

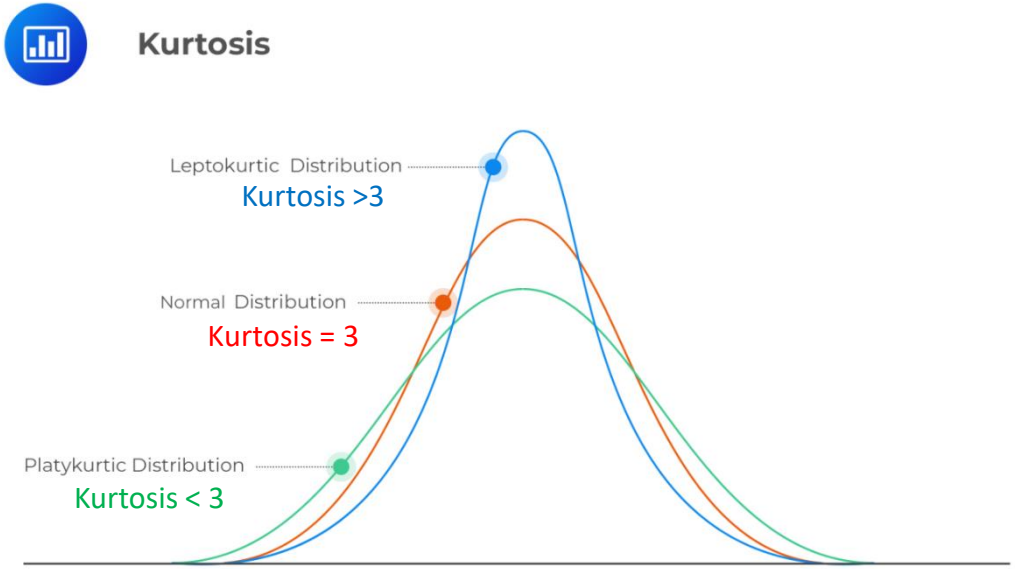
The Thumb rule

- If the skewness is between -0.5 to +0.5 then we can say data is fairly symmetrical.
- If the skewness is between -1 to -0.5 or 0.5 to 1 then data is moderately skewed.
- And if the skewness is less than -1 or greater than +1 then our data is heavily skewed.

- Heavily skewed data could be due to outliers.

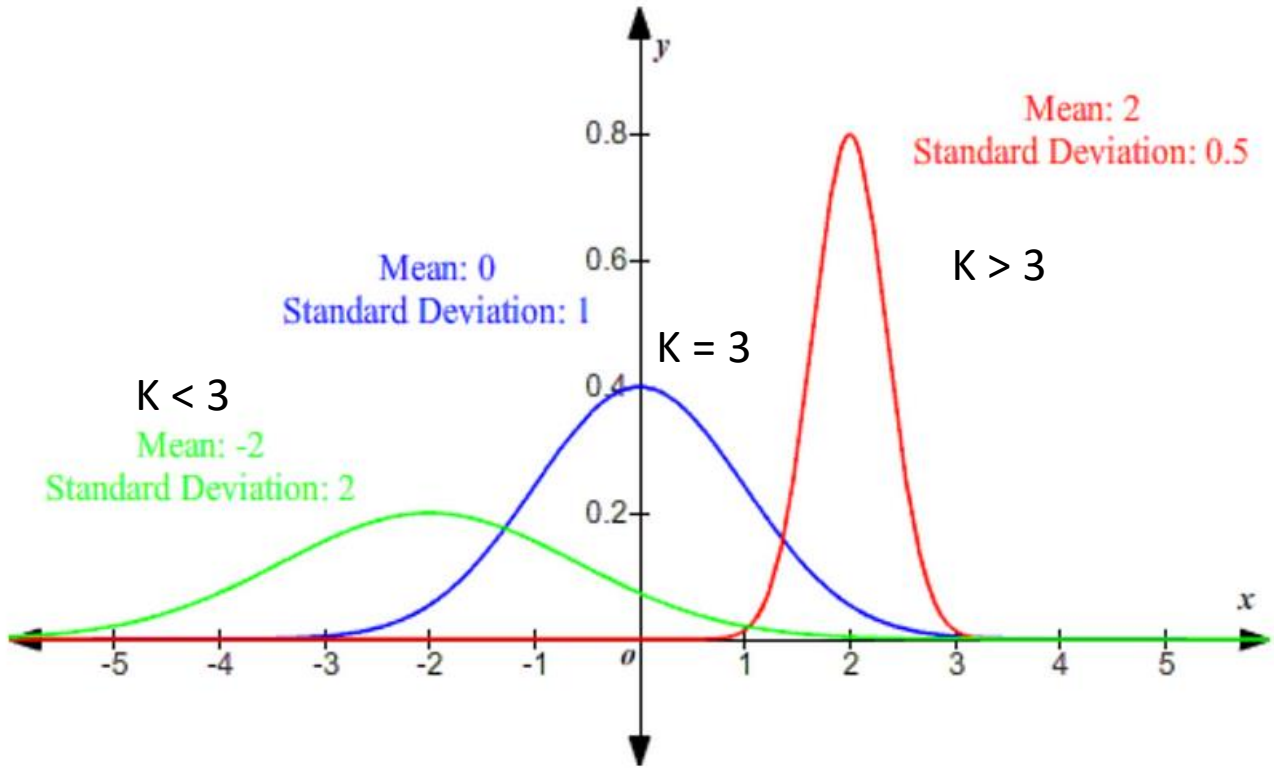
Kurtosis

- Kurtosis tells us about the shape or peakedness or flatness of the distribution
- Kurtosis tell us about the tails behaviour. It is actually the measure of outliers present in the distribution.



Interpretation of Kurtosis

- Smaller the Standard Deviation, Steeper the Distribution whereas Higher the Standard Deviation, Flatter the distribution.



Python..

Skewness

# skewness along the index axis across all columns
df.skew(axis = 0, skipna = True)

Fisher's definition (subtracts 3)

```
data = [88, 85, 82, 97, 67, 77, 74, 86, 81, 95, 77, 88, 85, 76, 81]

#calculate sample skewness
skew(data, bias=False)

0.032697

#calculate sample kurtosis
kurtosis(data, bias=False)

0.118157
```

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Topic 4: Business Insight Extraction

Statistical analysis and Business insights

- Statistical analysis is the **engine** that turns raw business data into **actionable insights**.
- The magic happens when we use the right statistical tools to answer the right business questions.

Identify the business context

- Statistical analysis is most powerful when tied to a specific decision.
- Examples:
  - “Why are sales dropping in the North region?”
  - “Which customers are most likely to churn?”
  - “What factors drive delivery delays?”
  - “Which marketing channel gives the best ROI?”
- Once the question is clear, the right statistical method becomes obvious.

Use Descriptive Statistics to understand what is happening

- This is the **first layer of insight**.
- We explore:
  - Averages (mean, median)
  - Variability (standard deviation, range)
  - Distribution shapes
  - Outliers

- **Business Value:** One quickly sees patterns like:
  - "Average order value is rising but variability is huge."
  - "One store is consistently underperforming."
  - "A small number of customers contribute 60% of revenue."
- This sets the stage for deeper analysis.

Use Diagnostic Analysis

- This is where relationships between variables matter.
- Tools include:
  - **Correlation** → Which factors move together
  - **Covariance** → Direction of movement
  - **Regression** → Which factors *drive* outcomes
  - **ANOVA** → Which groups differ significantly

- **Business Value:** We uncover drivers and root causes:
  - "Delivery time increases mainly due to warehouse processing delays."
  - "Discounts increase sales only for new customers."
  - "Customer churn is strongly linked to service response time

Use Predictive Statistics to forecast what will happen

- Once you understand patterns, you can predict future outcomes.
- Methods:
  - Time-series forecasting
  - Predictive regression models
  - Probability models
  - Survival analysis (for churn)
- **Business value:** You can plan ahead:
  - “Demand will spike 18% next quarter.”
  - “High-value customers have a 30% churn probability.”
  - “Inventory will run out in 12 day

Use prescriptive statistics to recommend what to do

- Here analytics becomes strategy.
- Techniques:
  - Optimization models
  - Scenario analysis
  - Simulation (e.g., Monte Carlo)
  - A/B testing

- **Business value:** You get actionable decisions:
  - "Increase staff by 2 during peak hours to reduce wait time by 40%."
  - "Marketing Channel B gives the highest ROI under all scenarios."
  - "A 5% price increase has minimal impact on demand."

Communicate insights in business language

- The final step is turning statistical findings into **clear, decision-ready insights**.
- Example transformation:
- **Statistical output:**
  Regression coefficient for delivery time = 0.42, $p < 0.01$
- **Business insight:**
  "Every 1-hour delay in warehouse processing increases total delivery time by 42 minutes. Fixing this bottleneck will reduce delays by 30–40%."
- This is where the statistical analysis shines — translating analytics into business clarity

How to use statistics for business insights

- Start with a business question
- Explore the data (descriptive)
- Find causes (diagnostic)
- Predict outcomes (predictive)
- Recommend actions (prescriptive)
- Communicate insights clearly

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Topic 5: Demos

Summary

- Understood Statistical Analysis techniques and their significance in inference.
- Studied Corelation techniques.
- Understood the distribution methods to determine skew, kurtosis and identification of outliers
- Elaborated the aspects of Understood Histograms, Box plots.
- Understood aspects of extracting business insights from Statistical Analyss
- Studied a few implementation demos.

Questions?